Systems Analysis and Control

Matthew M. Peet
Arizona State University

Lecture 18: The Frequency Response

Overview

In this Lecture, you will learn:

Introduction to the Frequency Domain

- Life without Newton
  - "Who needs a model, anyway?"
- Black Boxes.

Frequency Response

- Predicting Frequency Response
- Using Frequency Response Data
- Bode Plots

What is "The Frequency Response"

What happens when I shake it? It depends on the frequency of the vibration.

Definition 1.

The **Frequency Response** is the *steady-state* output of a system with sinusoidal input.

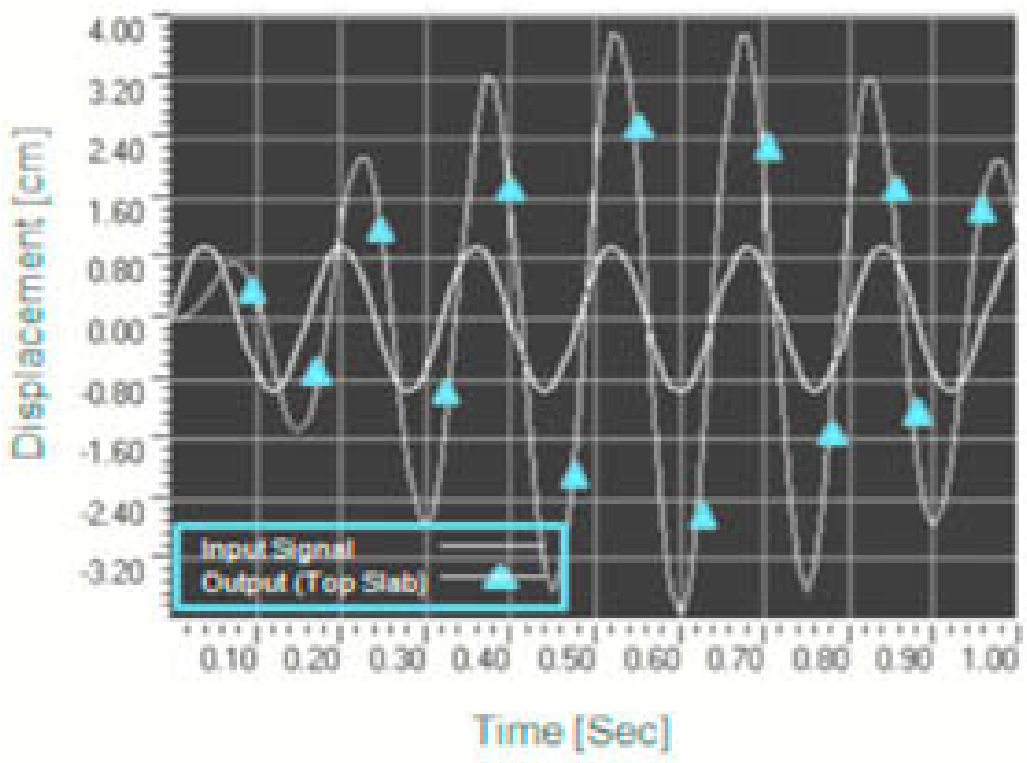


Figure: Response of Concrete Slabs to Soil Excitation (FEM)

The Frequency Response: Sinusoidal Inputs

Represent the input using Magnitude and Phase

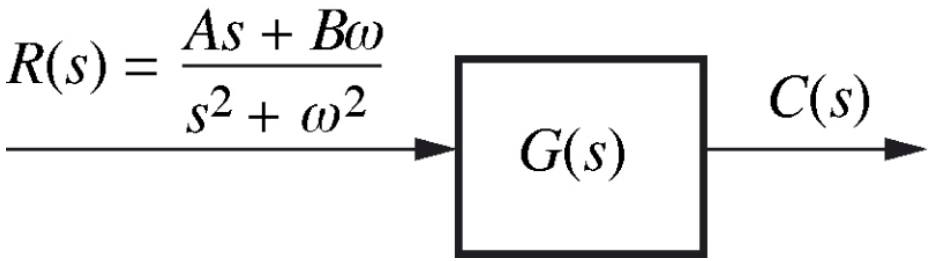
Recall: A Generic Sinusoidal Input

$$\begin{aligned} u(t) &= A\sin(\omega t) + B\cos(\omega t) \\ &= \sqrt{A^2+B^2}\sin\left(\omega t - \tan^{-1}\left(\frac{B}{A}\right)\right) \\ &= M\sin(\omega t + \phi) \end{aligned}$$

- $M = \sqrt{A^2+B^2}$
- $\phi = -\tan^{-1}\left(\frac{B}{A}\right)$

Laplace Transform (and Table):

$$\hat{u}(s) = \frac{Bs + A\omega}{s^2+\omega^2} = \frac{M\omega e^{\frac{\phi}{\omega}s}}{s^2+\omega^2}$$

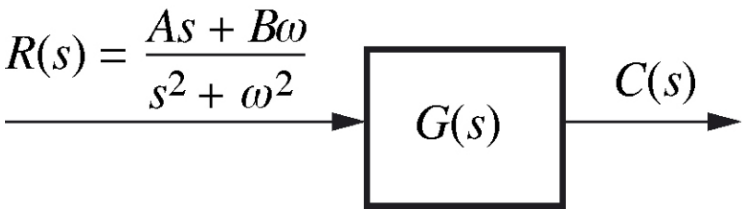


The Frequency Response: Sinusoidal Inputs

Apply the Partial Fraction Expansion to Obtain Output

For a given stable, strictly proper transfer function,

$$G(s) = \frac{n(s)}{(s+p_1)\cdots(s+p_n)},$$



then by partial-fraction expansion

$$\begin{aligned}\hat{y}(s) &= G(s)\hat{u}(s) \\ &= G(s)\frac{M\omega e^{\frac{\phi}{\omega}s}}{(s+\imath\omega)(s-\imath\omega)} \\ &= \frac{r_1}{s+p_1} + \cdots + \frac{r_n}{s+p_n} + \frac{\alpha}{s+\imath\omega} + \frac{\beta}{s-\imath\omega}.\end{aligned}$$

Finding the Steady-State Output for a Sinusoidal Input

Use Partial Fractions Expansion and Discard Transient Terms

Partial Fraction Expansion: $\hat{y}(s) = G(s)\hat{u}(s)$

$$\hat{y}(s) = \frac{r_1}{s+p_1} + \cdots + \frac{r_n}{s+p_n} + \frac{\alpha}{s+\imath\omega} + \frac{\beta}{s-\imath\omega}$$

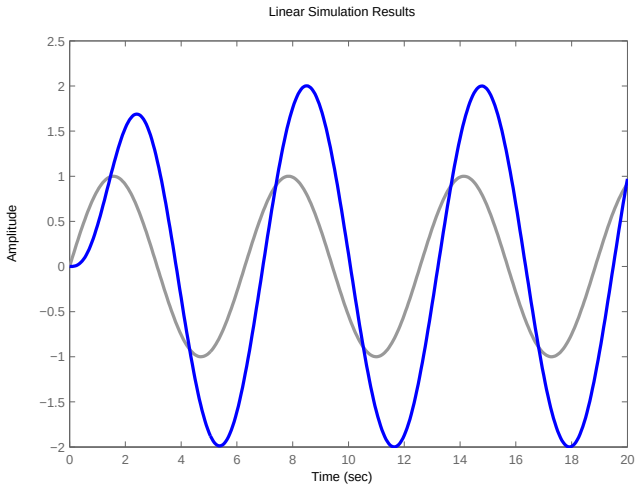
Inverse Laplace Transform:

$$y(t) = r_1 e^{-p_1 t} + \cdots + r_n e^{-p_n t} + \alpha e^{-\imath\omega t} + \beta e^{\imath\omega t}$$

But we want the **Steady-State Response**.

- Poles p_i are all stable.
  - $\lim_{t\to\infty} e^{-p_i t} = 0$
- These are called **Transient Responses**
- only left with

$$y_{ss}(t) = \alpha e^{-\imath\omega t} + \beta e^{\imath\omega t}$$



Apply Remainder Theorem to Partial Fraction Expansion

Since $\pm\imath\omega$ are isolated poles, by the remainder theorem:

$$\begin{aligned}\alpha &= G(s)\frac{M\omega e^{\frac{\phi}{\omega}s}}{(s+\imath\omega)(s-\imath\omega)}(s+\imath\omega)|_{s=-\imath\omega} \\ &= G(-\imath\omega)\frac{M\omega e^{-\imath\phi}}{-2\imath\omega} \\ &= G(-\imath\omega)\frac{Me^{-\imath\phi}}{-2\imath}\end{aligned}$$

Likewise,

$$\beta = G(\imath\omega)\frac{Me^{\imath\phi}}{2\imath}$$

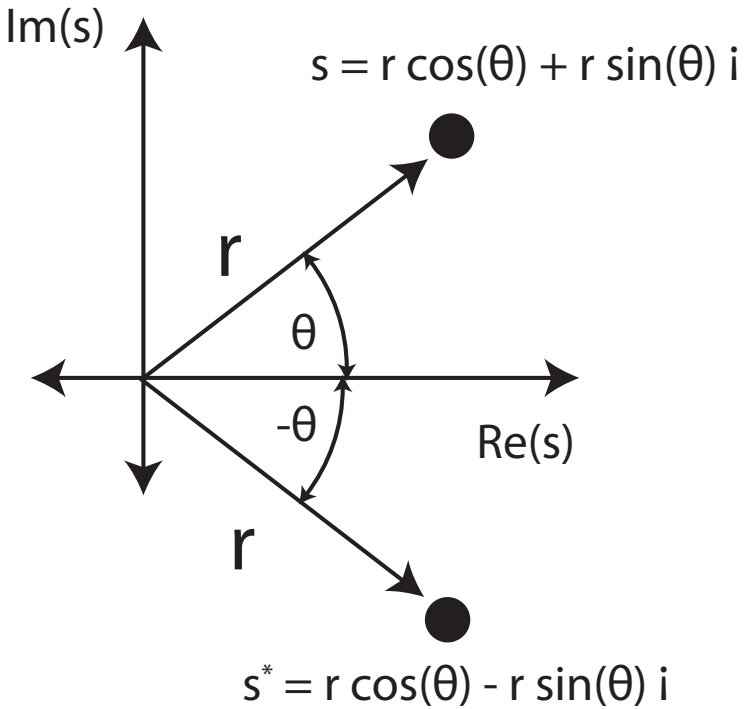
Then

$$\begin{aligned}y_{ss}(t) &= \alpha e^{-\imath\omega t} + \beta e^{\imath\omega t} \\ &= M\frac{G(\imath\omega)e^{\imath\omega t}e^{\phi\imath} - G(-\imath\omega)e^{-\imath\omega t}e^{\phi\imath}}{2\imath} \\ &= M\frac{G(\imath\omega)e^{\imath(\omega t+\phi)} - G(-\imath\omega)e^{-\imath(\omega t+\phi)}}{2\imath}\end{aligned}$$

Complex Numbers and Complex Conjugates

Complex Conjugate in polar form reverses the phase

Note: $G(-\imath\omega)$ is the complex conjugate of $G(\imath\omega)$.



Definition 2.

For a complex number $s = a + b\imath$, the **Complex Conjugate** of s is

$$s^* = a - b\imath$$

- Just replace $\imath \to -\imath$.
- $re^{\imath\theta} \to re^{-\imath\theta}$

Magnitude is unchanged. Phase is reversed

For $s = re^{\imath\theta}$,

Phase: $\angle s = \theta$

- $\angle s^* = -\theta = -\angle s$

Magnitude: $|s| = r$

- $|s^*| = r = |s|$

Simplifying the Steady-State Output

Using Polar Form of $G(\imath\omega) = |G(\imath\omega)|e^{\angle G(\imath\omega)\imath}$ and Euler's Formula

Recall the steady-state response

$$y_{ss}(t) = M\frac{G(\imath\omega)e^{\imath(\omega t+\phi)} - G(-\imath\omega)e^{-\imath(\omega t+\phi)}}{2\imath}$$

Now express $G(\imath\omega)$ as

$$G(\imath\omega) = |G(\imath\omega)|e^{\angle G(\imath\omega)\imath}$$

where $|G(\imath\omega)| = |G(\imath\omega)^*| = |G(-\imath\omega)|$, $\angle G(-\imath\omega) = \angle G(-\imath\omega)^* = -\angle G(\imath\omega)$

$$\begin{aligned}
y_{ss}(t) &= M\frac{G(\imath\omega)e^{\imath(\omega t+\phi)} - G(-\imath\omega)e^{-\imath(\omega t+\phi)}}{2\imath} \\
&= M|G(\imath\omega)|\frac{e^{\angle G(\imath\omega)}e^{\imath(\omega t+\phi)} - e^{-\angle G(\imath\omega)}e^{-\imath(\omega t+\phi)}}{2\imath} \\
&= M|G(\imath\omega)|\frac{e^{(\omega t+\phi+\angle G(\imath\omega))\imath} - e^{-(\omega t+\phi+\angle G(\imath\omega))\imath}}{2\imath} \\
&= M|G(\imath\omega)|\sin(\omega t + \phi + \angle G(\imath\omega))
\end{aligned}$$

Where we used Euler's formula: $\sin\theta = \frac{e^{\theta i} - e^{-\theta i}}{2\imath}$

Formula for Steady-State Frequency Response

Sinusoidal Input implies Sinusoidal Output (With Amplification and Phase Shift)

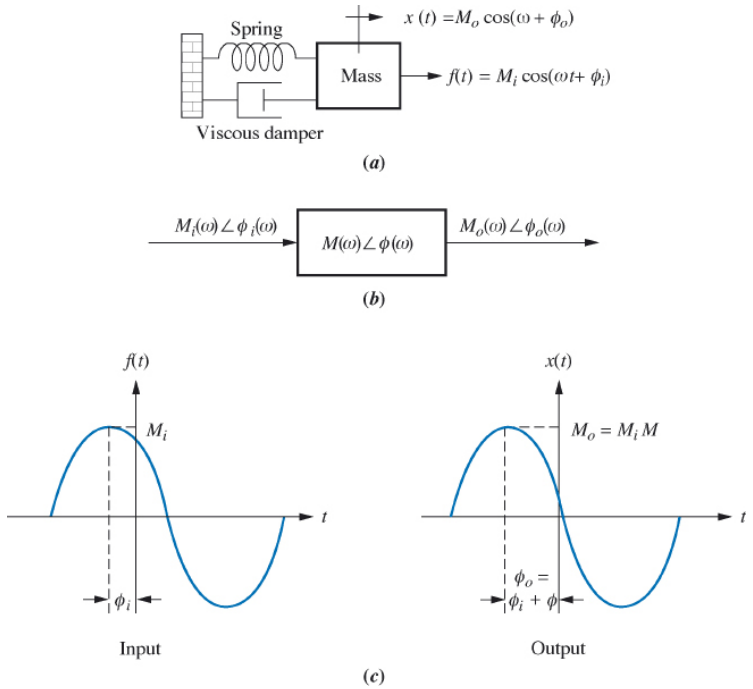
Input:

$$u(t) = M\sin(\omega t + \phi)$$

Output:

$$y(t) = M|G(\imath\omega)|\sin(\omega t + \phi + \angle G(\imath\omega))$$

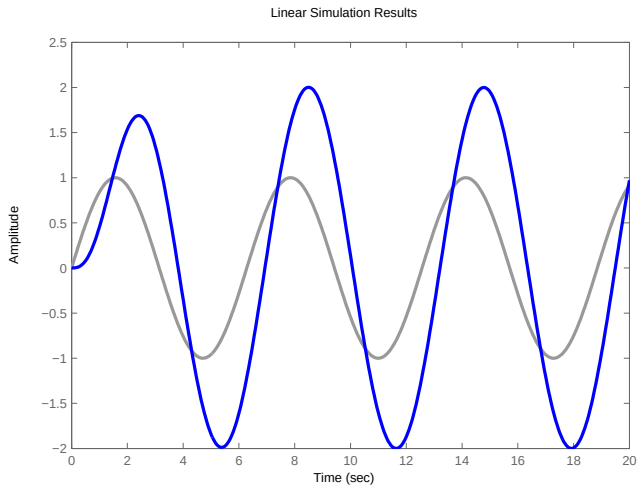
- Amplified by factor $|G(\imath\omega)|$
- Phase-shifted by angle $\angle G(\imath\omega)$
- Frequency is UNCHANGED!



The Frequency Response

Conclusion: The response to a sinusoidal input $\sin \omega t$:

- A sinusoid with the same frequency.
- *Phase* is shifted by $\angle G(\imath\omega)$.
- *Magnitude* is changed $|G(\imath\omega)|$.



We refer to

- $|G(\imath\omega)|$ is the Magnitude of Frequency Response
- $\angle G(\imath\omega)$ is the Phase of Frequency Response

These depend only on ω and $G(\imath\omega)$.

Complex Poles and Resonance

Large Amplification if Damping is Small

The amplification factor $|G(\imath\omega_n)|$ at the natural frequency, ω_n, is called resonance.

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \qquad |G(\imath\omega_n)| = \frac{1}{2\zeta}$$

If damping ratio is small, $\zeta \cong 0$, amplification is very large.

Figure: Frequency Sweeping with Resonance

Frequency Response Planning

Applications

Application: Crane Oscillation

- Sinusoidal Input from Hanging load.
- Avoid Spillage.
- Avoid Tipping.

A Form of **Motion Control**.

Frequency Response Planning

Applications

Figure: Simple Crane Sway Control

Figure: Industrial Crane Sway Control

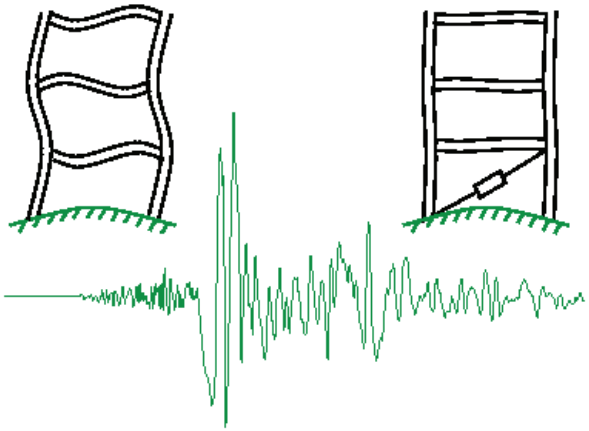
Figure: Failure of Crane Control

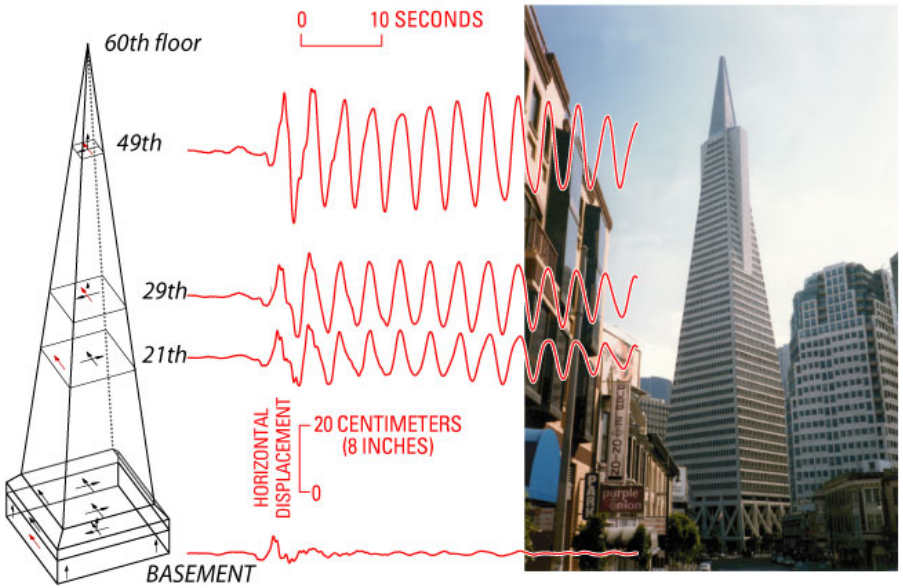
Frequency Response Planning

Modeling Structural Dynamics

Application: Building Response to Earthquakes

- Sinusoidal input from ground.
- Reduce peak output.





Obtaining Frequency Response Data

Controlling Structural Dynamics

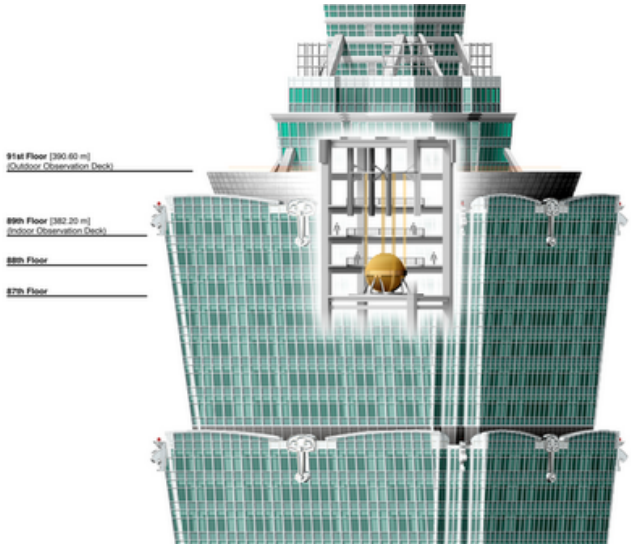


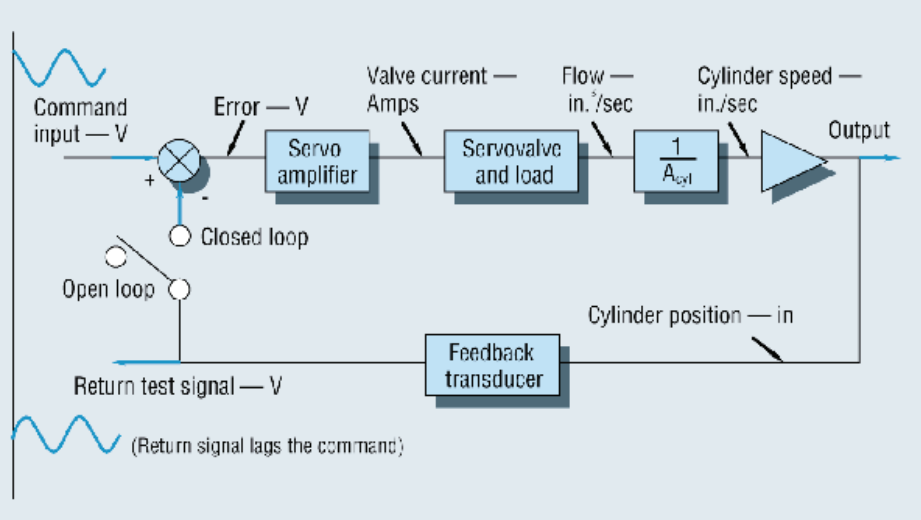
Figure: Earthquake Damping

Transfer Functions from Frequency Sweeping Data

No Model Required

This can work the other way too:

- Input $u(t) \sin \omega t$
- Output: $y(t) = M \sin(\omega t + \phi)$
- Measure M and ϕ
  - Relative Phase $\phi = \angle G(\imath\omega)$
  - Magnitude: $M = |G(\imath\omega)|$

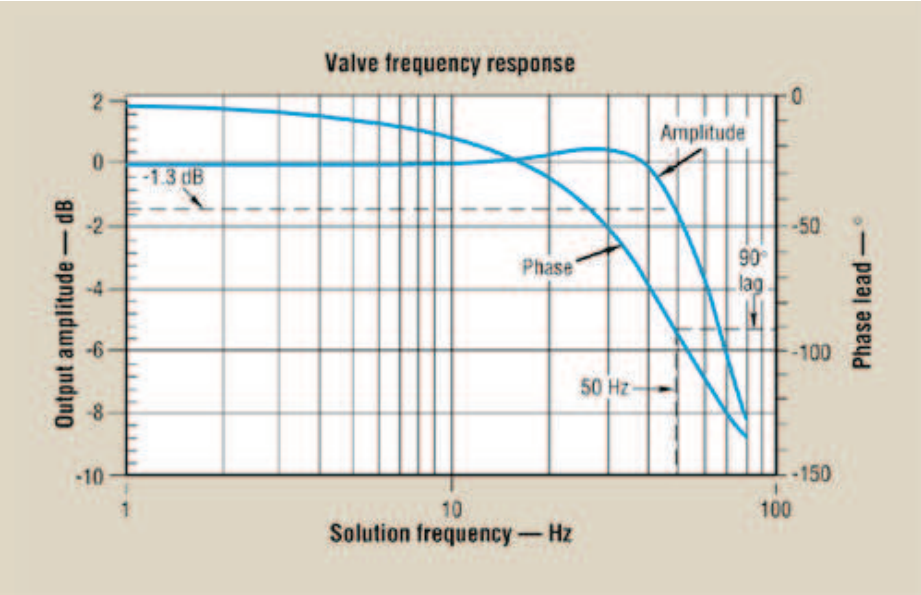


Frequency Sweeping: Measure M and ϕ at every frequency

- Get functions $M(\omega)$ and $\phi(\omega)$

Reconstruct

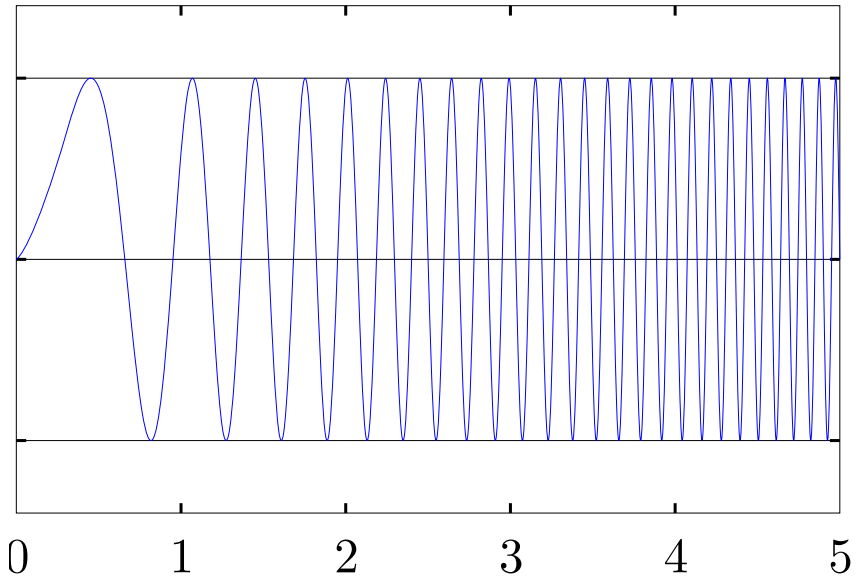
$$G(s) \cong M(s) e^{\phi(s)\imath}$$



The Chirp Signal used in Frequency Sweeping

Input: A Sinusoid of Increasing Frequency.

$$u(t) = \sin\left((\omega_0 + kt)t\right)$$



Complex Poles and Zeros

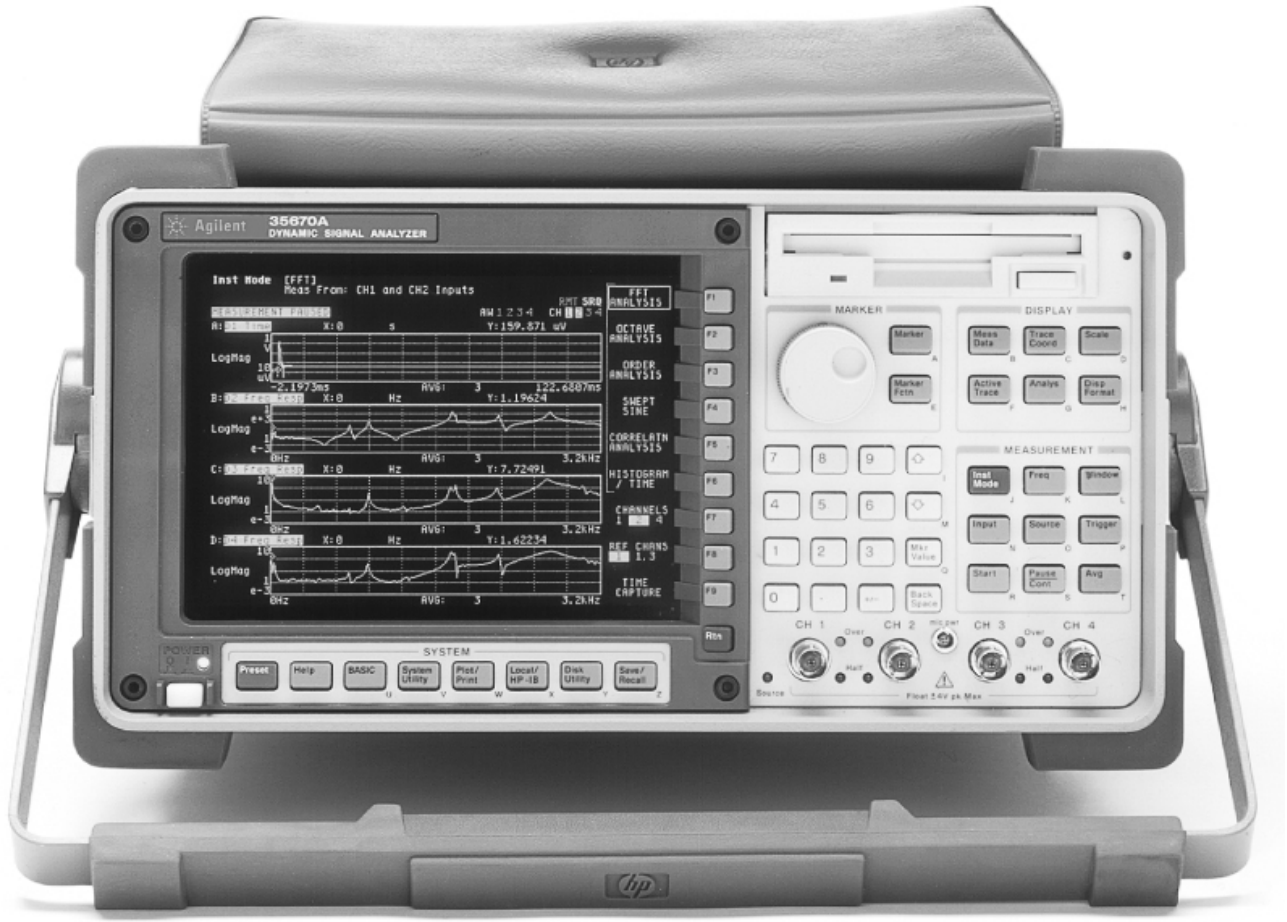
Figure: Frequency Sweeping with Resonance

The Frequency Response

Figure: A Frequency Sweep in Circuit Analysis

Frequency Sweeping Data From Oscilloscopes

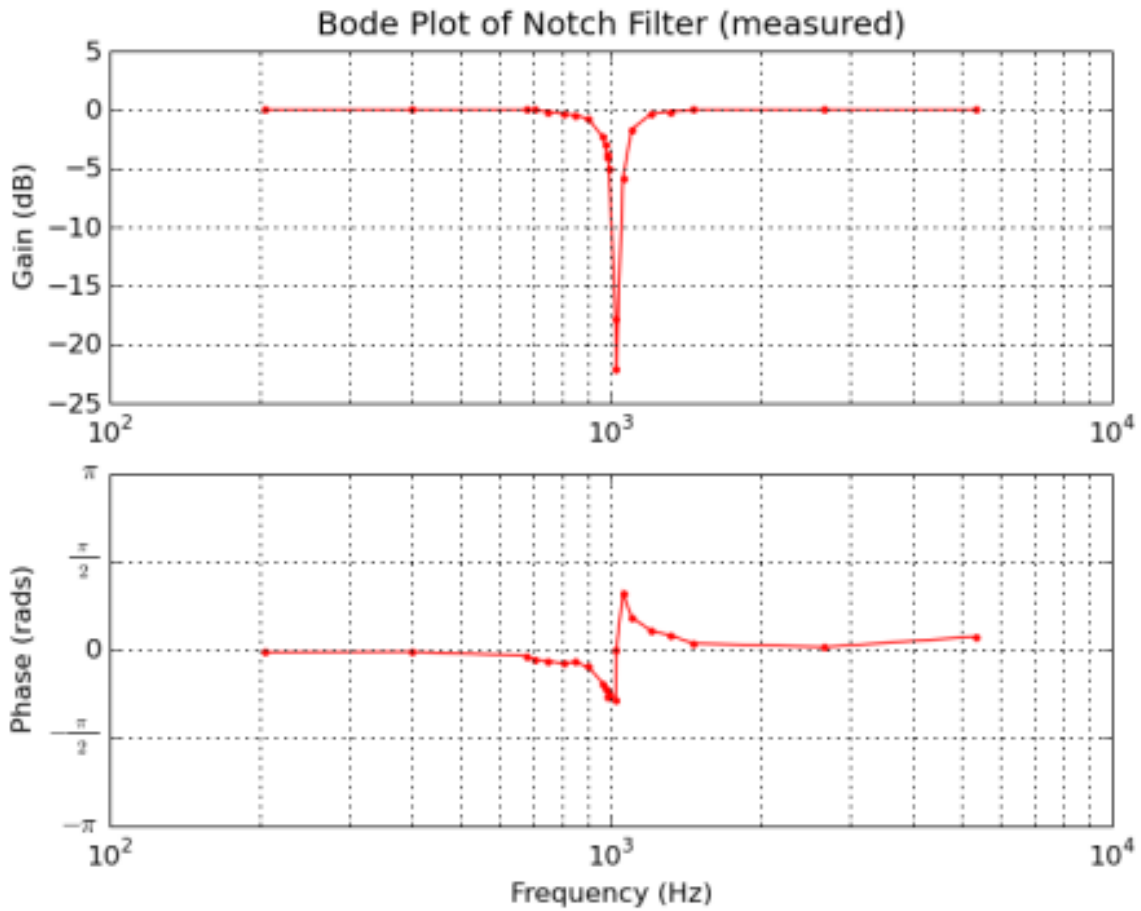
Magnitude and Phase Data



Visualizing Frequency Sweeping Data

Plotting Magnitude $|G(\imath\omega)|$ and Phase Changes $\angle G(\imath\omega)$ vs. Frequency, ω

Magnitude and Phase Data for a Notch Filter



This type of Magnitude-Phase graph is called a Bode Plot

Visualizing Frequency Sweeping Data

Plotting Magnitude $|G(\imath\omega)|$ and Phase Changes $\angle G(\imath\omega)$ vs. Frequency, ω

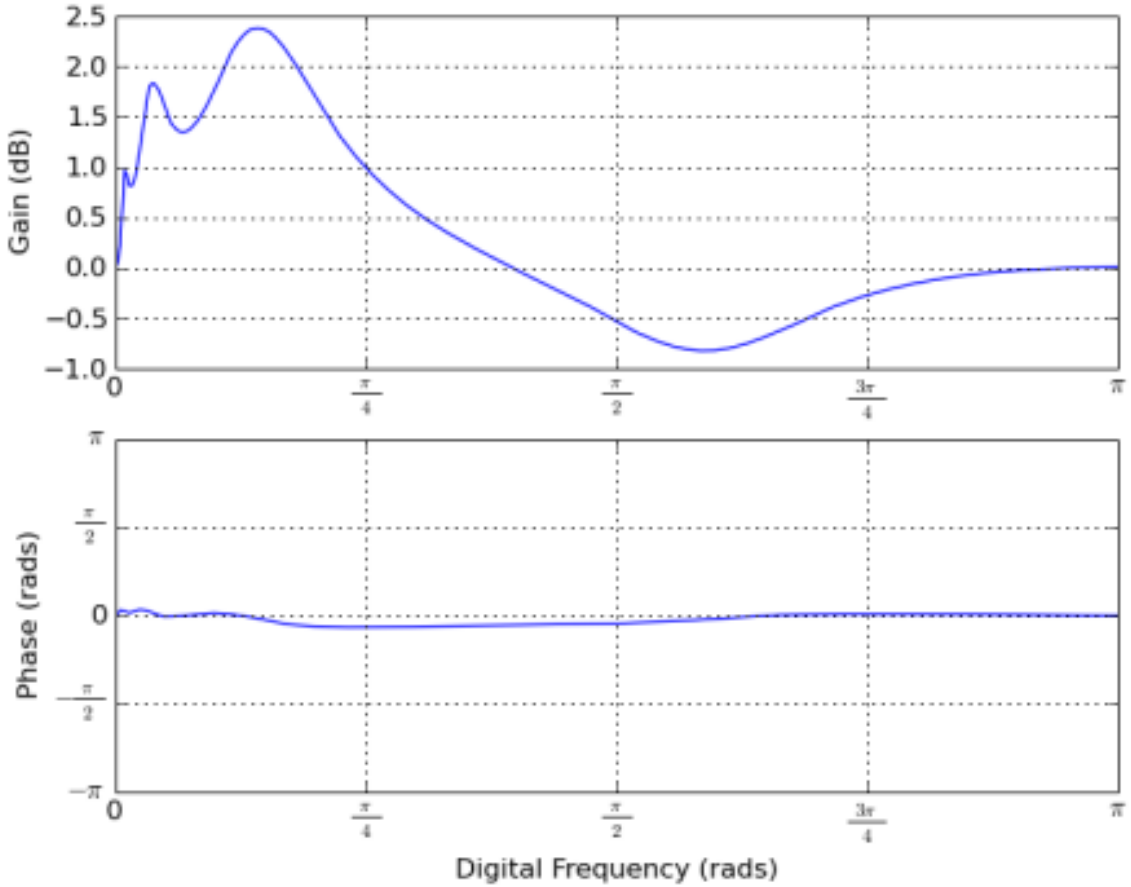


Figure: Data From a Graphic Equalizer

No Model is Required to understand the system.

Obtaining Frequency Response Data

Finite-Element Solvers Frequently Provide Frequency Response Data

For structures and rigid bodies.

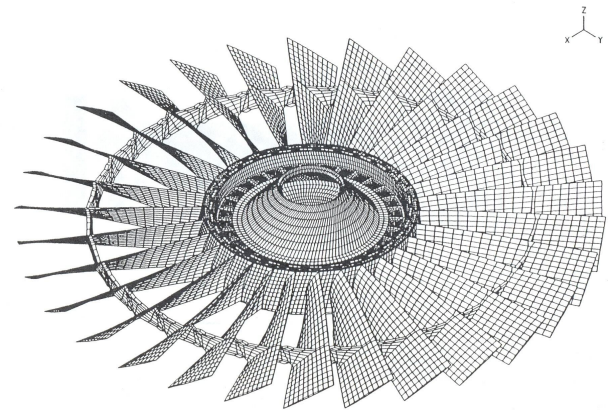
- Dynamics are Partial-Differential Equations
  - Elasticity
- We can derive the model, but it would be too complicated.

We must rely on Simulation.

- Simulate a sinusoidal input
  - Record output displacement
- Resulting model is only an approximation.

FEM/FEA Software which provide frequency response data.

- ANSYS
- Abaqus
- OptiStruct
- NASTRAN



Obtaining Frequency Response Data

Finite-Element Modeling

Figure: Satellite Frequency Response Analysis using NASTRAN

Summary

What have we learned today?

Introduction to the Frequency Domain

- Life without Newton
  - "Who needs a model, anyway?"
- Black Boxes.

Frequency Response

- Predicting Frequency Response
- Using Frequency Response Data
- Bode Plots

Next Lecture: The Bode Plot

Obtaining Frequency Response Data

Experimental Methods: Circuit Sweeping

Figure: Frequency Response Analysis in the Power Industry (Ad)